



Filed, Not Forgotten: A Field Study of Blame Recurrence After a Share House Adopts a Robot-Vacuum Incident Register

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Abstract. A shared robot vacuum accumulates a shared history of small blame — a sock ingested, a rug bunched against a doorway, a startled cat — and some share houses respond by keeping a written incident Register: a shared note or document logging what happened, when, and who was nearby. This paper reports a field study of 34 share houses recruited via co-op housing noticeboards and share-house classifieds, tracked over a 14-week term via a weekly two-minute check-in (431 of 476 possible house-weeks completed, 90.5%). Twenty-three houses maintained a Register at some point in the window; eleven never did. Register houses and No-Register houses argue about the vacuum at statistically indistinguishable rates (mean 0.59 versus 0.54 arguments per house-week; Mann-Whitney U , $p = 0.71$), and an ablation dropping Register status from a simple weekly-argument model costs it well under a percentage point of R^2 . What differs sharply is what an argument invokes: 61% of arguments in Register houses cite a specific prior logged entry, against 14% in houses with no register to cite (χ^2 , $p < 0.001$). Keeping a record of the vacuum's mishaps does not appear to settle them; it appears mainly to give the next argument something to point to.

Introduction

A household that shares a robot vacuum accumulates, alongside the vacuum's actual cleaning record, a second and less charted history: who was blamed for what it did. The vacuum eats a phone charger left on the floor, bunches a rug against a doorway, startles a cat mid-nap, or misses a room entirely on the day guests arrive, and somebody is nearby when it happens, or is assumed to be. In enough houses, this history eventually gets written down. A shared note, a page in a household document, or a running thread in the group chat titled something like “vacuum incidents” becomes the place a mishap is logged: what happened, roughly when, and who, if anyone, was in the room.

The households that adopt this practice tend to describe it, when asked, as a way of settling arguments rather than starting them — a record that lets a dispute resolve against what actually happened instead of against two competing

memories of it. Whether a written incident Register does what its adopters intend is, formally, an empirical question. Register-keeping is a well-studied institutional reflex in higher-stakes domains, but its capacity to change the behaviour it documents, rather than merely the record of that behaviour, has not previously been asked of arguments about a shared appliance, kept by the people living with it.

This paper reports a field study of 34 share houses tracked over a single 14-week term, comparing houses that keep a written vacuum-incident Register against houses that do not. Its contributions, each modest by design:

1. We compare vacuum-related argument frequency between Register and No-Register houses across a 14-week window, tracked via a weekly resident check-in.
2. We show Register houses' arguments differ sharply in what they invoke — a specific prior logged entry rather than the incident

at hand — without differing in how often an argument happens.

3. We report an ablation in which dropping Register-adoption status from a simple weekly-argument model changes its fit by under a percentage point, consistent with the group-level comparison.

The Register, on this evidence, is not shown to make arguments better or worse; it is shown to change what they cite.

Related work

Formal documentation of a disputed process, and the informal account given by the people subject to it, are a long-running comparative pairing in adjacent literatures. Roommate conflict research finds that people in shared housing consistently misattribute a co-resident's irritating behaviour to disposition while attributing their own equivalent behaviour to circumstance, a pattern first traced to roommate disputes directly [1] and since extended to how socioeconomic status shapes which conflict-resolution mode a roommate pair actually uses [2]. Shared housing carries a measurable wellbeing cost tied less to the arrangement than to how its frictions are handled [3], and satisfaction in shared single-person households tracks a space's perceived procedural fairness more reliably than its rules' content [4]; fairness judgement more broadly weights earned entitlement and self-interest asymmetrically [5].

A second, adjacent literature asks whether formally documenting an incident changes what happens after it. Healthcare incident-reporting is the most heavily studied version: practitioners value reporting in principle [6], but longitudinal audits of large reporting databases find the feedback loop from a logged incident back to preventive action chronically underused, with most logged cases producing no downstream recommendation at all [7], [8], [9]. Multi-user smart-home systems face a similar problem in miniature: conflict-resolution mechanisms for shared connected devices assume logging a device's action is the hard part, when whose account of it is accepted is often the actual site of dispute [10]. Domestic robots specifically acquire a personified, blame-bearing status in the households that live with them, with failure events shaping perceived trustworthiness independent of the failure's actual cause [11], a

status owners extend into how they imagine the robot's idle time as a household resource in its own right [12], continuous with earlier work on how a domestic robot's entry renegotiates existing chore expectations among the people who live there [13], [14]. Self-monitoring interventions offer the closest general precedent for what a Register might do: they reliably produce a small but real shift in the behaviour being logged [15], a baseline this paper's own households, on the evidence below, do not meet.

Closer to home, prior work at this institution traces the same gap: an energy-budget power-board's published fairness rotation predicts less than a quarter of which resident's appliance is actually cut once load rank is available [16], and a household chore wheel's posted assignment converges with its logged completion only after an appeals process is introduced, with underlying task-completion holding flat throughout [17]. Both findings share a shape with the one below: a documented mechanism changes what a household can point to, not what it does.

Method

Sites and instrumentation

Thirty-four share houses running a single shared robot vacuum in daily use were recruited via co-op housing noticeboards and share-house classifieds and tracked over one 14-week term. Each house completed a two-minute weekly check-in naming any vacuum-related incident from the week (what happened, and who, if anyone, was nearby) and any argument that incident, or an earlier one, prompted. Response across the term totalled 431 of a possible 476 house-weeks (90.5%). A house was classified as a Register house if, at any point in the window, it kept a written incident Register — a shared note, document, or whiteboard logging vacuum mishaps for later reference — and a No-Register house otherwise; 23 houses fell into the first group and 11 into the second, a split set entirely by what households were already doing before the study rather than by any intervention this paper introduced.

Incident and argument coding

Two coders independently classified each of the 288 self-reported incidents into one of four categories — property damage, navigation obstruction, startled pet or person, and missed-area or perceived shirking — reconciled after a pilot pass, and classified each of the 214 self-reported arguments by what it invoked: a specific prior incident, the current incident only, or general, unspecified blame with no incident named. For Register houses, an invoked prior incident was checked where possible against the house’s own Register text (21 of 23 houses’ logs were sighted at least once); for No-Register houses, an invoked prior incident could only be a remembered one, since no logged entry existed to name.

Comparison and ablation

We define each house h ’s citation rate as

$$C_h = \frac{A_h}{T_h},$$

where A_h is the number of house h ’s arguments that invoke a specific prior incident and T_h is house h ’s total argument count, and compare citation rate between Register and No-Register houses with a chi-squared test on the pooled three-category classification above. We compare mean weekly argument rate per house between groups via a two-sided Mann-Whitney U test, and, to test whether Register status carries predictive weight beyond what the group comparison already suggests, fit a minimal linear model of mean weekly argument rate against Register status, household size, and the vacuum’s months in service, refitting it with Register status removed and comparing R^2 across the two specifications.

Results



Figure 1: What a vacuum-related argument invokes, by whether the house keeps a written incident Register. Register houses’ arguments cite a specific prior logged entry 61% of the time ($n = 158$); No-Register houses’ arguments do so 14% of the time ($n = 56$), relying on the current incident or unspecified blame instead.

The composition difference in Figure 1 is large and consistent with the citation-rate definition above: pooled across the 23 Register houses, mean $C_h = 0.58$; across the 11 No-Register houses, mean $C_h = 0.13$ (χ^2 , $p < 0.001$). Register houses do not argue about the vacuum more — they argue with more of the house’s own history on the table, echoing findings on multi-user smart-home conflict tools, where logging a device’s action turns out to be the easy part and whose account of it is accepted remains the actual site of dispute [10].



Figure 2: Mean vacuum-related arguments per house-week, Register houses ($n = 23$) against No-Register houses ($n = 11$). The distributions overlap almost completely across the 14-week window.

Figure 2 shows the frequency side finds no such separation: mean weekly argument rate is 0.59 in Register houses and 0.54 in No-Register houses, distributions overlapping across nearly

their full range, with no reliable difference between them (Mann-Whitney U , $p = 0.71$).

Incident category	Share of 288 logged incidents
Property damage (cord, sock, toy)	34%
Navigation obstruction	27%
Startled pet or person	21%
Missed area / perceived shirking	18%

The incident mix itself does not differ meaningfully between Register and No-Register houses (not shown); what differs is only what a house does with the mix once an argument starts. The ablation changes almost nothing: a minimal model of weekly argument rate against Register status, household size, and vacuum months-in-service explains a modest share of variance ($R^2 = 0.112$); dropping Register status alone costs the model 0.6 percentage points of R^2 (n.s.), consistent with the group-level comparison above. We retain Register status in the reported model for completeness, and because no other variable in the check-in data plausibly stands in for it.

Discussion

Two independent measures agree on where a Register's effect sits and where it does not. It does not sit in argument frequency: the two groups argue about their shared vacuum at rates a two-sided test cannot separate, and an ablation removing Register status from a simple frequency model costs it almost nothing. It sits, instead, in what an argument invokes once it starts: a Register house's disputes draw overwhelmingly on a specific, nameable prior incident, where a No-Register house's disputes fall back on the incident at hand or on unspecified blame.

This is not the outcome the households who adopt a Register typically describe wanting. A record kept to settle disputes against what actually happened instead functions, on this evidence, mainly as material the next dispute can cite — closer to the chronically underused feed-

back loop documented in healthcare incident reporting [7], [8], [9] than to the self-monitoring literature's modest behaviour-change effect [15]. The vacuum's mishaps get filed. What happens next to the household arguing about them does not appear to depend on the filing.

Limitations

Register adoption was observed, not assigned: houses that already keep a shared incident log may differ systematically from houses that do not, in ways this single-term, 34-house sample cannot fully separate from the Register's own effect, though the frequency result's consistency across the group comparison and the ablation suggests the effect, if any, is small. Argument and incident counts are self-reported rather than independently observed, which protects against the intrusiveness a household-installed recording device would introduce but leaves the categories dependent on what a house was willing to name on a weekly form. The 14-week window covers one term; whether a Register's citation effect persists or fades over a longer tenancy remains open.

Conclusion

A written incident Register does not appear to change how often a share house argues about its robot vacuum; it changes what the argument is about. Across 34 houses tracked for one term, Register houses' disputes cite a specific prior logged incident at more than four times the rate of houses with no register to cite, while the two groups' weekly argument rates remain statistically indistinguishable. Filing an incident, on this evidence, is not the same act as resolving it. Future work might interview the households who keep a Register about what, if anything, they believe a cited entry is supposed to accomplish once it is raised in an argument — a question this study, built from the weekly outside view of 34 households, was not designed to put to the people actually reaching for the log mid-dispute.

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